SECTION INDEX: Cotter & Knuckle Joints Classification Index

Machine Element / Joint Type	Primary Design Parameter / Objective	Databook Section
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Sleeve and Cotter Joint	Design 60 kN sleeve joint (d, d_2, t, d_1, b, a, c), double shear & crushing checks	Section 2
Gib and Cotter Joint	Design 50 kN connecting rod big end joint ($B_1, t, t_1, t_3, B, b_1, b, t_4, l_1, l_2$)	Section 3
Knuckle Joint (Standard)	Design 150 kN knuckle joint ($d, d_1, d_2, d_3, t, t_1, t_2$), 8 stress checks	Section 4
Knuckle Joint (with FOS)	Design 70 kN tie rod knuckle joint with FOS = 6 (d, d_1, d_2, d_3, t, t_1), 4 stress checks	Section 5

Design Standard Note: *All design equations, shear failures, crushing stress limits, and empirical proportion checks conform strictly to standard mechanical design principles (R.S. Khurmi & J.K. Gupta).*

SECTION 1: Socket and Spigot Cotter Joint (30 kN Load)

System Parameters

Tension / Compression Load: $P = 30 \text{ kN} = 30000 \text{ N}$

Carbon Steel Stresses: Tensile/Compressive $\sigma_t = \sigma_c = 50 \text{ MPa}$, Shear $\tau = 35 \text{ MPa}$, Crushing $\sigma_c = 90 \text{ MPa}$

Design Protocol

Step 1: Size rod diameter d in tension

Step 2: Size spigot diameter d_2 & cotter thickness t (Trial & Crushing Check)

Step 3: Size socket outer diameter d_1, cotter width b, socket collar d_4, c, rod end a, & spigot collar d_3, t_1

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1. Rod Diameter (d)

Tensile failure of rod

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$30000 = \frac{\pi}{4} d^2 (50) = 39.3 d^2$$

$$d^2 = \frac{30000}{39.3} = 763$$

$$d = 27.6 \text{ mm}$$

$$\Rightarrow \boldsymbol{d = 28 \text{ mm}} \quad (\textbf{Ans.})$$

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2. Trial Spigot Diameter (d_2) & Cotter Thickness (t)

Tension failure of spigot across slot ($t = \frac{d_2}{4}$)

$$P = \left[\frac{\pi}{4}d_2^2 - d_2 t\right]\sigma_t$$

$$30000 = \left[\frac{\pi}{4}d_2^2 - d_2\left(\frac{d_2}{4}\right)\right]50 = 26.8(d_2)^2$$

$$(d_2)^2 = \frac{30000}{26.8} = 1119.4$$

$$d_2 = 33.4 \text{ mm} \Rightarrow 34 \text{ mm}, \quad t = \frac{34}{4} = 8.5 \text{ mm}$$

3. Crushing Stress Check & Recalculation (d_2, t)

Check crushing stress for trial dimensions

$$\sigma_c = \frac{P}{d_2 \cdot t} = \frac{30000}{34 \times 8.5} = 103.8\text{N/mm}^2 > 90 \text{ MPa} \quad (\textbf{UNSAFE}$$

$$30000 = d_2\left(\frac{d_2}{4}\right)90 = 22.5(d_2)^2$$

$$(d_2)^2 = \frac{30000}{22.5} = 1333$$

$$d_2 = 40 \text{ mm} \quad (\textbf{Ans.})$$

$$t = \frac{40}{4} = \textbf{10 mm} \quad (\textbf{Ans.})$$

4. Socket Outer Diameter (d_1)

Tension failure of socket across slot

$$P = \left[\frac{\pi}{4}(d_1^2 - d_2^2) - (d_1 - d_2)t\right]\sigma_t$$

$$30000 = \left[\frac{\pi}{4}(d_1^2 - 40^2) - (d_1 - 40)(10)\right]50$$

$$0.7854d_1^2 - 10d_1 - 1854.6 = 0$$

$$d_1 = 49.9 \text{ mm}$$

$$\Rightarrow \boldsymbol{d_1 = 50 \text{ mm}} \quad (\textbf{Ans.})$$

5. Width of Cotter (b)

Double shear failure of cotter

$$P = 2b \cdot t \cdot \tau$$

$$30000 = 2b \times 10 \times 35 = 700b$$

$$b = \frac{30000}{700} = \mathbf{43 \text{ mm}} \quad (\textbf{Ans.})$$

6. Socket Collar Diameter (d_4)

Crushing failure of socket collar

$$P = (d_4 - d_2)t \cdot \sigma_c$$

$$30000 = (d_4 - 40) \times 10 \times 90 = 900(d_4 - 40)$$

$$d_4 - 40 = \frac{30000}{900} = 33.3$$

$$d_4 = 73.3 \text{ mm}$$

$$\Rightarrow \boldsymbol{d_4 = 75 \text{ mm}} \quad (\textbf{Ans.})$$

7. Socket Collar Thickness (c)

Double shear failure of socket end

$$P = 2(d_4 - d_2)c \cdot \tau$$

$$30000 = 2(75 - 40)c \times 35 = 2450c$$

$$c = \frac{30000}{2450} = \mathbf{12\ mm} \quad (\mathbf{Ans.})$$

8. Slot End to Rod End Distance (a)

Double shear failure of rod end

$$P = 2a \cdot d_2 \cdot \tau$$

$$30000 = 2a \times 40 \times 35 = 2800a$$

$$a = \frac{30000}{2800} = 10.7 \text{ mm}$$

$$\Rightarrow \boldsymbol{a} = \mathbf{11\ mm} \quad (\mathbf{Ans.})$$

9. Spigot Collar Diameter (d_3)

Crushing failure of spigot collar

$$P = \frac{\pi}{4}(d_3^2 - d_2^2)\sigma_c$$

$$30000 = \frac{\pi}{4}(d_3^2 - 40^2)90$$

$$d_3^2 - 1600 = 424$$

$$d_3 = \sqrt{2024} = \mathbf{45\ mm} \quad (\mathbf{Ans.})$$

10. Spigot Collar Thickness (t_1)

Shear failure of spigot collar

$$P = \pi d_2 \cdot t_1 \cdot \tau$$

$$30000 = \pi \times 40 \times t_1 \times 35 = 4400 t_1$$

$$t_1 = \frac{30000}{4400} = 6.8 \text{ mm}$$

$$\Rightarrow \boldsymbol{t_1 = 8 \text{ mm}} \quad (\textbf{Ans.})$$

11. Cotter Length (l) & Clearance Dimension (e)

Empirical length and clearance

$$l = 4d = 4 \times 28 = \mathbf{112 \text{ mm}} \quad (\textbf{Ans.})$$

$$e = 1.2d = 1.2 \times 28 = 33.6 \text{ mm}$$

$$\Rightarrow \boldsymbol{e = 34 \text{ mm}} \quad (\textbf{Ans.})$$

12. Complete Design Output

Socket & Spigot Joint Summary: d = 28 mm, d2 = 40 mm, t = 10 mm, d1 = 50 mm, b = 43 mm, d4 = 75 mm, c = 12 mm, a = 11 mm, d3 = 45 mm, t1 = 8 mm, l = 112 mm, e = 34 mm

SECTION 1: Socket and Spigot Cotter Joint Assembly — MECHANICAL DIAGRAM

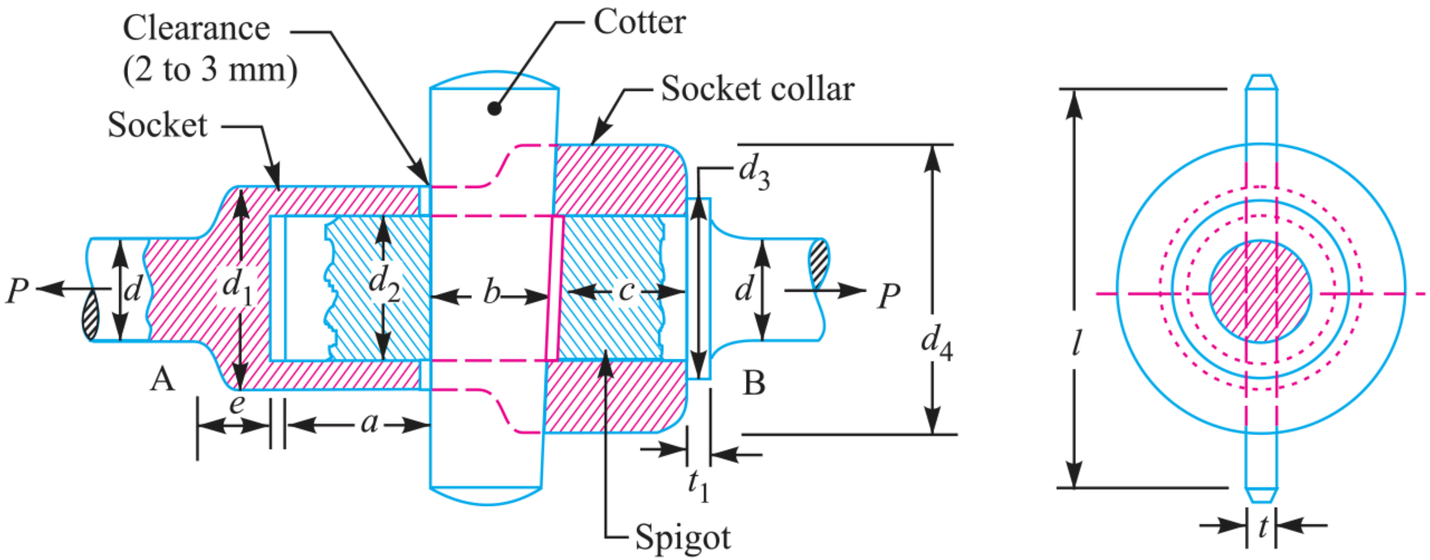

Fig. 12.1. Socket and spigot cotter joint.

Figure 12.1: Socket & Spigot Cotter Joint Assembly & Proportions

SECTION 2: Sleeve and Cotter Joint (60 kN Load)

System Parameters

Tensile Load: $P = 60 \text{ kN} = 60000 \text{ N}$

Allowable Stresses: Tensile $\sigma_t = 60 \text{ MPa}$, Shear $\tau = 70 \text{ MPa}$, Crushing $\sigma_c = 125 \text{ MPa}$

Design Protocol

Step 1: Size rod diameter d in tension

Step 2: Size enlarged rod end d_2 & cotter thickness t ($t = \frac{d_2}{4}$) with crushing check

Step 3: Size sleeve outer diameter d_1, cotter width b, rod shear distance a, & sleeve shear distance c

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1. Rod Diameter (d)

Tensile failure of rod

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$60000 = \frac{\pi}{4} d^2 (60) = 47.13 d^2$$

$$d^2 = \frac{60000}{47.13} = 1273$$

$$d = 35.7 \text{ mm}$$

$$\Rightarrow \boldsymbol{d = 36 \text{ mm}} \quad (\textbf{Ans.})$$

2. Enlarged Rod Diameter (d_2) & Cotter Thickness (t)

Tension failure of rod across slot ($t = \frac{d_2}{4}$)

$$P = \left[\frac{\pi}{4}d_2^2 - d_2 t\right]\sigma_t$$

$$60000 = \left[\frac{\pi}{4}d_2^2 - d_2\left(\frac{d_2}{4}\right)\right]60 = 32.13(d_2)^2$$

$$(d_2)^2 = \frac{60000}{32.13} = 1867$$

$$d_2 = 43.2 \text{ mm}$$

$$\Rightarrow \boldsymbol{d_2 = 44 \text{ mm}} \quad (\textbf{Ans.})$$

$$t = \frac{44}{4} = \textbf{11 mm} \quad (\textbf{Ans.})$$

3. Crushing Stress Check

Check crushing stress on rod end / cotter

$$\sigma_c = \frac{P}{d_2 \cdot t}$$

$$= \frac{60000}{44 \times 11}$$

$$= \mathbf{124 N/mm^2} \leq 125 \text{ MPa} \quad (\textbf{SAFE})$$

4. Sleeve Outer Diameter (d_1)

Tension failure of sleeve across slot

$$P = \left[\frac{\pi}{4}(d_1^2 - d_2^2) - (d_1 - d_2)t\right]\sigma_t$$

$$60000 = \left[\frac{\pi}{4}(d_1^2 - 44^2) - (d_1 - 44)(11)\right]60$$

$$0.7854d_1^2 - 11d_1 - 1036.7 = 0$$

$$d_1 = 58.4 \text{ mm}$$

$$\Rightarrow \boldsymbol{d_1 = 60 \text{ mm}} \quad (\textbf{Ans.})$$

5. Width of Cotter (b)

Double shear failure of cotter

$$P = 2b \cdot t \cdot \tau$$

$$60000 = 2b \times 11 \times 70 = 1540b$$

$$b = \frac{60000}{1540} = 38.96 \text{ mm}$$

$$\Rightarrow \boldsymbol{b = 40 \text{ mm}} \quad (\textbf{Ans.})$$

6. Rod End Shear Distance (a)

Double shear failure of rod end

$$P = 2a \cdot d_2 \cdot \tau$$

$$60000 = 2a \times 44 \times 70 = 6160a$$

$$a = \frac{60000}{6160} = 9.74 \text{ mm}$$

$$\Rightarrow \boldsymbol{a = 10 \text{ mm}} \quad (\textbf{Ans.})$$

7. Sleeve End Shear Distance (c)

Double shear failure of sleeve end

$$P = 2(d_1 - d_2)c \cdot \tau$$

$$60000 = 2(60 - 44)c \times 70 = 2240c$$

$$c = 26.78 \text{ mm}$$

$$\Rightarrow \boldsymbol{c = 28 \text{ mm}} \quad (\textbf{Ans.})$$

8. Complete Design Output

Sleeve & Cotter Joint Summary: d = 36 mm, d2 = 44 mm, t = 11 mm, d1 = 60 mm, b = 40 mm, a = 10 mm, c = 28 mm

SECTION 2: Sleeve and Cotter Joint Assembly — MECHANICAL DIAGRAM

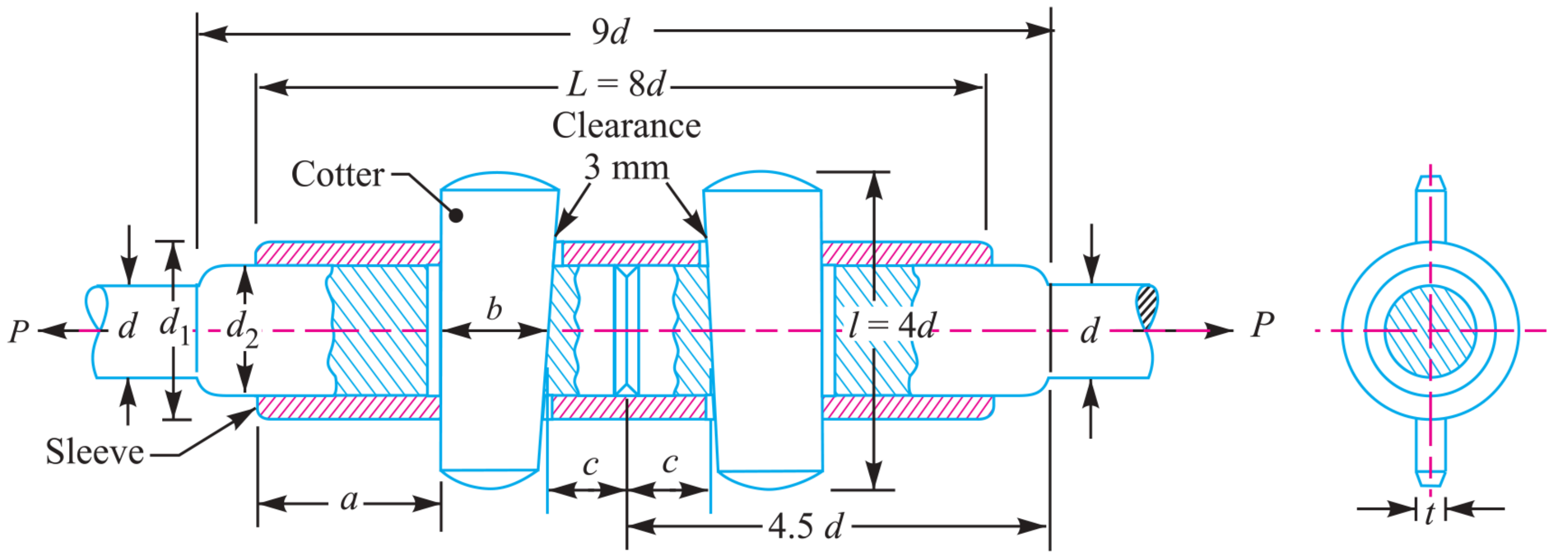

Fig. 12.9. Sleeve and cotter joint.

Figure 12.9: Sleeve & Cotter Joint Assembly & Proportions

SECTION 3: Gib and Cotter Joint for Connecting Rod Big End (50 kN Load)

System Parameters

Max Load: $P = 50 \text{ kN} = 50000 \text{ N}$ | Rod Diameter Adjacent to Strap: $d = 75 \text{ mm}$

Strap Tensile Stress: $\sigma_t = 25 \text{ MPa}$ | Cotter & Gib Shear Stress: $\tau = 20 \text{ MPa}$

Design Protocol

Step 1: Determine strap width $B_1 = d = 75 \text{ mm}$, cotter/gib thickness $t = \frac{B_1}{4} = 20 \text{ mm}$

Step 2: Size strap thinnest section t_1, strap thickness at cotter t_3, & total width B (divided into gib b_1 & cotter b)

Step 3: Determine strap end t_4, rod end l_1, & overhang l_2

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1. Strap Width (B_1) & Cotter/Gib Thickness (t)

Width equal to rod diameter $d = 75 \text{ mm}$

$$B_1 = d = \mathbf{75 \text{ mm}} \quad (\mathbf{Ans.})$$

$$t = \frac{B_1}{4} = \frac{75}{4} = 18.75 \text{ mm}$$

$$\Rightarrow \boldsymbol{t} = \mathbf{20 \text{ mm}} \quad (\mathbf{Ans.})$$

2. Thinnest Strap Thickness (t_1)

Tensile failure of strap at thinnest section

$$P = 2B_1 \cdot t_1 \cdot \sigma_t$$

$$50000 = 2 \times 75 \times t_1 \times 25 = 3750t_1$$

$$t_1 = \frac{50000}{3750} = 13.3 \text{ mm}$$

$$\Rightarrow \boldsymbol{t_1 = 15 \text{ mm}} \quad (\textbf{Ans.})$$

3. Strap Thickness at Cotter Hole (t_3)

Equal cross-sectional area criterion

$$2t_3(B_1 - t) = 2t_1B_1$$

$$2t_3(75 - 20) = 2 \times 15 \times 75$$

$$110t_3 = 2250$$

$$t_3 = \frac{2250}{110} = 20.45 \text{ mm}$$

$$\Rightarrow \boldsymbol{t_3 = 21 \text{ mm}} \quad (\textbf{Ans.})$$

4. Total Width of Gib & Cotter (B)

Double shear failure of gib and cotter

$$P = 2B \cdot t \cdot \tau$$

$$50000 = 2B \times 20 \times 20 = 800B$$

$$B = \frac{50000}{800} = 62.5 \text{ mm}$$

$$\Rightarrow \boldsymbol{B = 65 \text{ mm}} \quad (\textbf{Ans.})$$

5. Division of Width between Gib (b_1) & Cotter (b)

Empirical proportions (55% gib, 45% cotter)

$$b_1 = 0.55B = 0.55 \times 65 = 35.75 \text{ mm}$$

$$\Rightarrow \boldsymbol{b_1 = 36 \text{ mm}} \quad (\textbf{Ans.})$$

$$b = 0.45B = 0.45 \times 65 = 29.25 \text{ mm}$$

$$\Rightarrow \boldsymbol{b = 30 \text{ mm}} \quad (\textbf{Ans.})$$

6. Empirical Proportions (t_4, l_1, l_2)

Strap end, rod end, and overhang dimensions

$$t_4 = 1.25t_1 = 1.25 \times 15 = 18.75 \text{ mm}$$

$$\Rightarrow \boldsymbol{t_4 = 20 \text{ mm}} \quad (\textbf{Ans.})$$

$$l_1 = 2t_1 = 2 \times 15 = \mathbf{30 \text{ mm}} \quad (\textbf{Ans.})$$

$$l_2 = 2.5t_1 = 2.5 \times 15 = 37.5 \text{ mm}$$

$$\Rightarrow \boldsymbol{l_2 = 40 \text{ mm}} \quad (\textbf{Ans.})$$

7. Complete Design Output

Gib & Cotter Joint Summary: B1 = 75 mm, t = 20 mm, t1 = 15 mm, t3 = 21 mm, B = 65 mm (Gib b1 = 36 mm, Cotter b = 30 mm), t4 = 20 mm, l1 = 30 mm, l2 = 40 mm

SECTION 3: Gib and Cotter Joint Assembly — MECHANICAL DIAGRAM

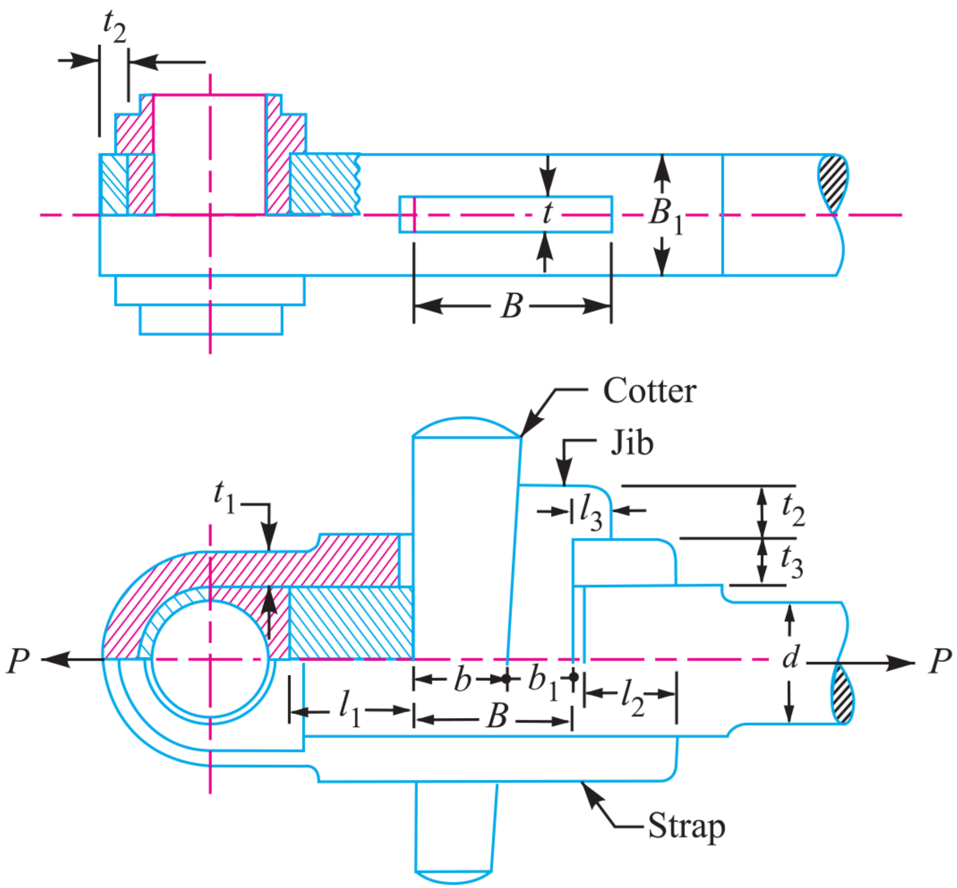

Fig. 12.12. Gib and cotter joint for strap end of a connecting rod.

Figure 12.12: Gib and Cotter Joint Construction for Connecting Rod Big End

SECTION 4: Knuckle Joint Design & Stress Checks (150 kN Load)

System Parameters

Tensile Load: $P = 150 \text{ kN} = 150000 \text{ N}$

Allowable Stresses: Tension $\sigma_t = 75 \text{ MPa}$, Shear $\tau = 60 \text{ MPa}$, Compression $\sigma_c = 150 \text{ MPa}$

Design Protocol

Step 1: Size rod diameter d in tension & calculate empirical dimensions $(d_1, d_2, d_3, t, t_1, t_2)$

Step 2: Verify pin shear, single eye tension/shear/crushing, & fork end tension/shear/crushing

1. Rod Diameter (d) & Empirical Proportions

Tensile failure of solid rod

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$150000 = \frac{\pi}{4} d^2 (75) = 59 d^2$$

$$d^2 = \frac{150000}{59} = 2540$$

$$d = \mathbf{52\ mm} \quad (\mathbf{Ans.})$$

$$d_1 = d = \mathbf{52\ mm}, \quad d_2 = 2d = \mathbf{104\ mm}$$

$$d_3 = 1.5d = \mathbf{78\ mm}, \quad t = 1.25d = \mathbf{65\ mm}$$

$$t_1 = 0.75d = \mathbf{40\ mm}, \quad t_2 = 0.5d = \mathbf{26\ mm}$$

2. Knuckle Pin Double Shear Check (τ)

Shear stress check on pin

$$P = 2\left(\frac{\pi}{4}d_1^2\right)\tau$$

$$150000 = 2\left(\frac{\pi}{4}\right)(52)^2\tau = 4248\tau$$

$$\tau = \frac{150000}{4248} = \mathbf{35.3\ MPa} \leq 60\ \text{MPa} \quad (\mathbf{SAFE})$$

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3. Single Eye Tension Check (σ_t)

Tearing across pin hole

$$P = (d_2 - d_1)t \cdot \sigma_t$$

$$150000 = (104 - 52)65 \cdot \sigma_t = 3380\sigma_t$$

$$\sigma_t = \frac{150000}{3380} = \mathbf{44.4\ MPa} \qquad \leq 75\ \text{MPa}$$

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4. Single Eye Shear Check (τ)

Double shearing of eye end

$$P = (d_2 - d_1)t \cdot \tau$$

$$150000 = (104 - 52)65 \cdot \tau = 3380\tau$$

$$\tau = \frac{150000}{3380} = \mathbf{44.4\ MPa} \qquad \leq 60\ \text{MPa} \quad (\mathbf{S}$$

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5. Single Eye Crushing Check (σ_c)

Crushing between pin and eye

$$P = d_1 \cdot t \cdot \sigma_c$$

$$150000 = 52 \times 65 \cdot \sigma_c = 3380\sigma_c$$

$$\sigma_c = \frac{150000}{3380} = \mathbf{44.4\ MPa} \leq 150\ \text{MPa} \quad (\mathbf{SAFE})$$

6. Fork End Tension Check (σ_t)

Tearing across fork pin holes

$$P = (d_2 - d_1)2t_1 \cdot \sigma_t$$

$$150000 = (104 - 52)2(40) \cdot \sigma_t = 4160\sigma_t$$

$$\sigma_t = \frac{150000}{4160} = \mathbf{36\ MPa} \qquad \leq 75\ \mathrm{N}$$

7. Fork End Shear Check (τ)

Double shearing of fork ends

$$P = (d_2 - d_1)2t_1 \cdot \tau$$

$$150000 = (104 - 52)2(40) \cdot \tau = 4160\tau$$

$$\tau = \frac{150000}{4160} = \mathbf{36\ MPa} \qquad \leq 60\ \mathrm{MP}$$

8. Fork End Crushing Check (σ_c)

Crushing between pin and fork

$$P = d_1 \cdot 2t_1 \cdot \sigma_c$$

$$150000 = 52 \times 2(40) \cdot \sigma_c = 4160\sigma_c$$

$$\sigma_c = \frac{150000}{4160} = \mathbf{36\ MPa} \qquad \leq 150\ \mathrm{MPa} \quad (\mathbf{SA}$$

9. Complete Design Output

Knuckle Joint Summary: d = 52 mm, d1 = 52 mm, d2 = 104 mm, d3 = 78 mm, t = 65 mm, t1 = 40 mm, t2 = 26 mm, All 8 Stress Checks SAFE

SECTION 4: Knuckle Joint Assembly — MECHANICAL DIAGRAM

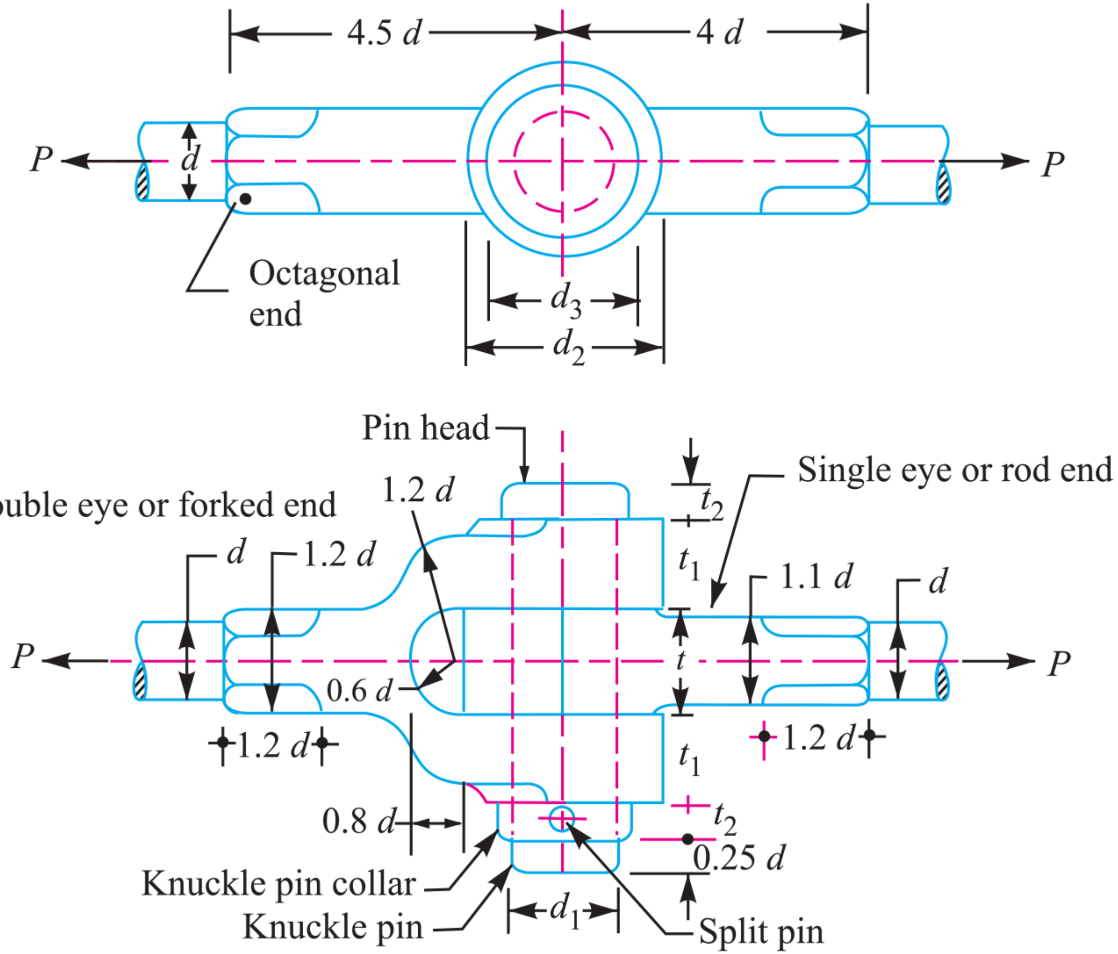

Fig. 12.16. Kunckle joint.

Figure 12.16: Knuckle Joint Construction & Pin Assembly Details

SECTION 5: Knuckle Joint Design with Factor of Safety (70 kN Load)

System Parameters

Pull Load: $P = 70\ \text{kN} = 70000\ \text{N}$ | Factor of Safety: FOS $= 6$

Rod Ultimate Strength: $\sigma_{\text{tu}} = 420\ \text{MPa} \Rightarrow \text{Permissible}\ \ \sigma_t = \frac{420}{6} = 70\text{N/mm}^2$

Pin Ultimate Shear Strength: $\tau_u = 396\ \text{MPa} \Rightarrow \text{Permissible}\ \ \tau = \frac{396}{6} = 66\text{N/mm}^2$

Design Protocol

Step 1: Size rod diameter d in tension & calculate empirical dimensions (d_1, d_2, d_3, t, t_1)

Step 2: Verify pin shear stress τ, single eye tension σ_t, & fork end tension σ_t

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1. Rod Diameter (d) & Empirical Proportions

Tensile failure of tie rod under allowable $\sigma_t = 70\ \text{MPa}$

$$P = \frac{\pi}{4} d^2 \cdot \sigma_t$$

$$70000 = \frac{\pi}{4} d^2 (70) = 55 d^2$$

$$d^2 = \frac{70000}{55} = 1273$$

$$d = \mathbf{36\ mm} \quad (\mathbf{Ans.})$$

$$d_1 = d = \mathbf{36\ mm}, \quad d_2 = 2d = \mathbf{72\ mm}$$

$$d_3 = 1.5d = \mathbf{54\ mm}, \quad t = 1.25d = \mathbf{45\ mm}$$

$$t_1 = 0.75d = \mathbf{27\ mm}$$

2. Knuckle Pin Double Shear Check (τ)

Check shear stress for allowable $\tau = 66\ \text{MPa}$

$$P = 2\left(\frac{\pi}{4}d_1^2\right)\tau$$

$$70000 = 2\left(\frac{\pi}{4}\right)(36)^2\tau = 2036\tau$$

$$\tau = \frac{70000}{2036} = \mathbf{34.4\ MPa}\ \le 66\ \text{MPa} \quad (\mathbf{SAFE})$$

3. Single Eye Tension Check (σ_t)

Check tensile stress for allowable $\sigma_t = 70\ \text{MPa}$

$$P = (d_2 - d_1)t \cdot \sigma_t$$

$$70000 = (72 - 36)45 \cdot \sigma_t = 1620\sigma_t$$

$$\sigma_t = \frac{70000}{1620} = \mathbf{43.2\ MPa} \qquad \le 70\ \text{MPa} \quad ($$

4. Fork End Tension Check (σ_t)

Check tensile stress for allowable $\sigma_t = 70\ \text{MPa}$

$$P = (d_2 - d_1)2t_1 \cdot \sigma_t$$

$$70000 = (72 - 36)2(27) \cdot \sigma_t = 1944\sigma_t$$

$$\sigma_t = \frac{70000}{1944} = \mathbf{36\ MPa} \qquad \le 70\ \text{M}$$

5. Complete Design Output

Knuckle Joint (FOS = 6) Summary: d = 36 mm, d1 = 36 mm, d2 = 72 mm, d3 = 54 mm, t = 45 mm, t1 = 27 mm, All Stress Checks SAFE

SECTION 5: Knuckle Joint Assembly — MECHANICAL DIAGRAM

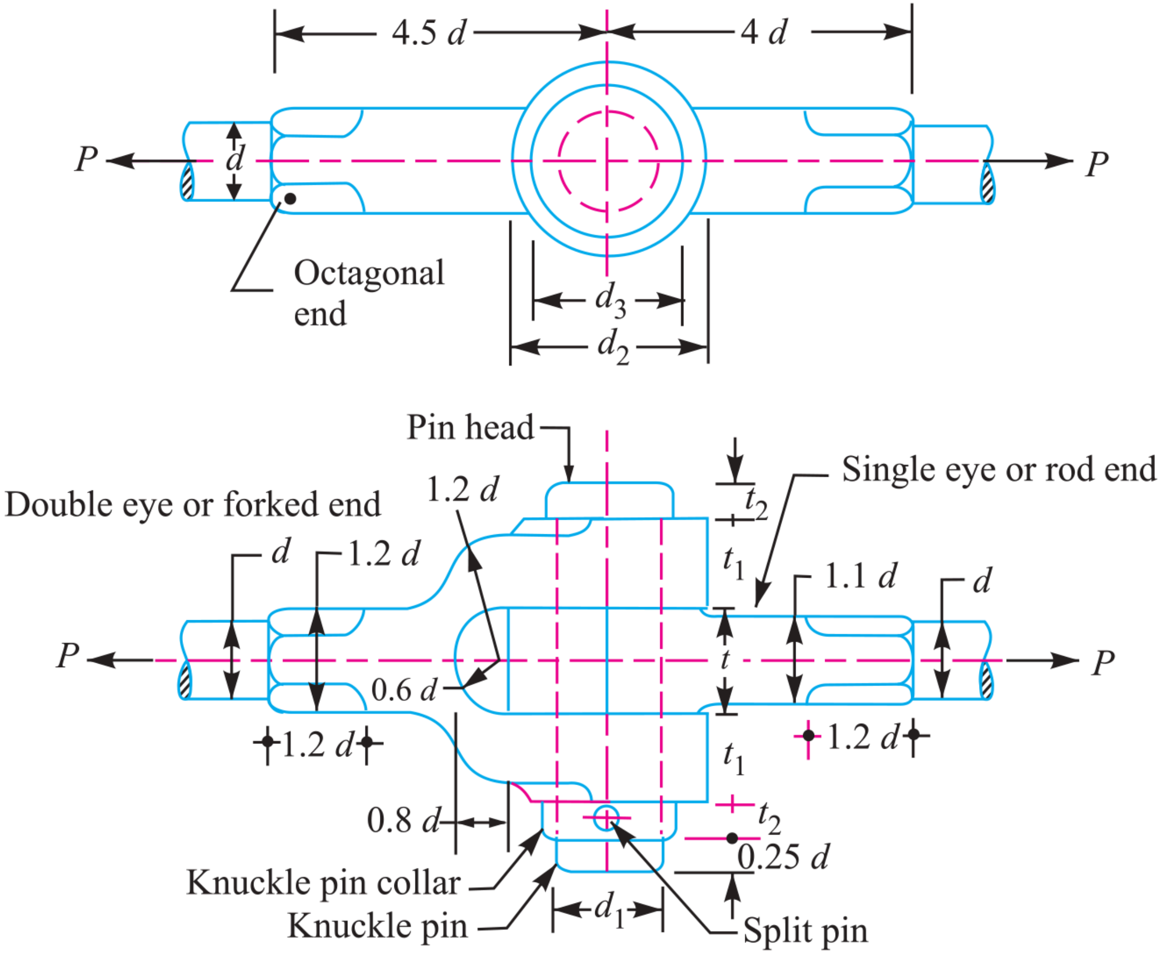

Fig. 12.16. Kunckle joint.

Figure 12.16: Knuckle Joint Fork & Pin Details